

BOLT, Base Paper and Panel Answer Sheet

1. Which Is Our Base Paper?

Ke, Z., Cao, Y., Chen, Z., Yin, Y., He, S., and Cheng, Y. (2025), “Early warning of cryptocurrency reversal risks via multi-source data,” Finance Research Letters, Article 107890, Elsevier.

This paper qualifies as a base paper, and it is worth stating why before the panel asks. It is an original empirical research article published in a Scopus-indexed, peer-reviewed Elsevier journal. It is not a review and it is not a survey. The authors build a model, train it on real data, report numerical results, and carry out ablation and robustness testing. This directly answers the objection raised at the previous review.

2. Which Type of Model, and Which Model, Was Used?

Table 1. Modelling choices in the base paper.

Aspect	What the base paper used
Task type	Binary classification of a rare risk event, not price regression
Primary model	LSTM (Long Short-Term Memory recurrent neural network)
Comparison models	Logistic Regression (Logit), Linear SVM, Random Forest
Reason for LSTM	To capture temporal dependencies across multi-source signals preceding the event
Imbalance handling	SMOTE (Synthetic Minority Over-sampling Technique)
Interpretability	SHAP attribution over the input feature set

The composition of that comparison set is significant. The base paper evaluates LSTM against Logistic Regression, SVM and Random Forest. It does not include XGBoost and it does not include GRU. That absence is one of our stated extensions, and it is a concrete, checkable gap rather than a vague claim of improvement.

3. On Which Dataset Was It Trained?

The base paper uses a multi-source dataset combining three families of signal, which is unusual in this literature and constitutes the paper’s principal claim to novelty. The first family is blockchain metrics, including hash rate and large-value or “whale” transactions. The second is social sentiment extracted from public discussion of cryptocurrency. The third is regulatory signals, treating regulatory announcements and policy events as shock variables.

The prediction target is a pin-bar reversal risk event in the Bitcoin market, a market-microstructure reversal pattern which the authors argue occurs frequently in Bitcoin and is uncommon in traditional assets.

Confirm the exact date range and observation count from the paper’s data section before the review. If asked and the figure is not to hand, say that the paper draws blockchain, sentiment and regulatory series for Bitcoin and that the precise window will be confirmed, never guess a number in front of a panel.

4. How Accurate Is the Base Paper’s Model?

Table 2. Results reported in the base paper.

Metric reported	Value
LSTM F1-score	0.703
Rolling-window F1 across market cycles	0.64 – 0.72
Contribution of blockchain features (SHAP)	greater than 33% of predictive power
Effect of SMOTE on rare-event recall	+19%

Two observations about these numbers strengthen our position rather than weaken it.

First, the paper reports F1 rather than accuracy. That is correct practice for a rare-event problem, and it is precisely why our own work refuses to quote accuracy. If the panel asks why accuracy is absent from our results, the answer is that the base paper omits it for the same reason: a model predicting “no event” every day would score above ninety-five per cent accuracy and provide no value whatsoever.

Second, an F1 of approximately 0.70 is a realistic and honest result rather than a spectacular one. This is the genuine difficulty level of the problem. If our system reports a figure in a similar range, that is a credible outcome. Conversely, any report of 0.95 or above on rare-event crash prediction should raise an immediate suspicion of information leakage.

5. Which Market Did the Base Paper Comment On?

The Bitcoin market specifically, a single asset, treated in isolation. The paper does not model cross-asset relationships, does not treat Ethereum or the wider altcoin market as interacting entities, and does not address cross-chain effects.

Critically, the authors themselves state that future research may investigate cross-chain dynamics and more advanced temporal models. Our extension is therefore not a criticism we invented; it is the direction the original authors identified as unfinished. Stating that we are continuing the line of work the authors proposed is a considerably stronger position than claiming to have found a flaw.

6. Our Dataset: Difference, Source and Authenticity

6.1 Where the data comes from

Table 3. Data sources and the basis on which each is authentic.

Source	What we take	Authenticity basis
CoinGecko API	Price, volume, market capitalisation, market dominance	Public API, widely used in published research, documented endpoints
Binance Public API	Historical OHLCV candlestick data	Data from the exchange where the trades actually occurred, a primary source, not an aggregator
Etherscan / Dune Analytics	Exchange inflows and outflows, whale wallet movements, active addresses	Read directly from the public blockchain ledger, cryptographically verifiable and impossible to falsify
CryptoPanic / news RSS archives	Timestamped news headlines	Published items carrying verifiable publication timestamps
Reddit Public API	Public community discussion	Public posts with API-provided timestamps

6.2 Did we make it ourselves?

We assemble and label the dataset ourselves; we do not fabricate any part of it. The distinction should be stated precisely: no private or proprietary data is collected, every raw observation originates from a public API or the public blockchain, and our contribution is the alignment, the point-in-time construction and the crash labelling. The entire dataset regenerates from our scripts, so any third party can rebuild it exactly and check it.

6.3 How it differs from the base paper

Table 4. Dataset comparison against the base paper.

Dimension	Base paper	BOLT
Assets	Bitcoin only	Multiple leading cryptocurrencies, treated as an interacting set
Target event	Pin-bar reversal (a candlestick microstructure pattern)	Severe drawdown beyond a defined threshold within a forward horizon
Market data	Focus on non-conventional signals	Full market data plus on-chain plus sentiment
Cross-asset structure	Not modelled	Correlation, lead-lag and contagion features across assets
Crisis coverage	Not organised around named crisis episodes	Deliberately spans Terra/LUNA, FTX and the USDC de-peg for validation
Reproducibility	Dataset not released as a benchmark	Regenerable from scripts, released with the labelling protocol

The strongest single difference to lead with is the target. The base paper predicts a technical chart pattern; we predict a severe market decline of defined magnitude. These are different targets with different economic meaning, a pin-bar reversal is a trading signal, whereas a crash is a systemic event. That is a substantive change of research question, not merely a change of dataset.

7. Which Neural Network Have We Used?

An LSTM as the core sequence model, with a GRU variant for comparison. The LSTM is the same architecture used in the base paper, chosen deliberately so that our results are directly comparable against a published benchmark rather than against nothing. The GRU is a simpler gated recurrent architecture with fewer parameters, included because with rare events and limited data a smaller model frequently generalises better, a comparison the base paper never performed.

The LSTM is implemented from first principles in NumPy. This provides direct access to internal gate activations, which is required for gradient-based attribution, and demonstrates command of the architecture rather than dependence on a framework abstraction.

8. Why LSTM and XGBoost, and Why These Models Only?

This question must be answered carefully, because asserting a preference for deep learning without justification is the weakest available response. The two models address different halves of the problem.

The LSTM handles the sequence. A crash is not signalled by a single variable crossing a threshold; it is signalled by a pattern developing across several days, in which volatility rises while exchange inflows accelerate and sentiment deteriorates. An LSTM reads a window in order and carries earlier context forward. A tabular model cannot represent this unless we hand-craft the temporal features ourselves, which requires knowing the answer in advance.

XGBoost handles the engineered tabular features. Gradient boosting is consistently strong on structured financial data, handles missing values and mixed scales natively, trains in seconds, and is highly compatible with SHAP attribution, on which our explainability layer depends. Several published comparisons find gradient boosting competitive with or superior to deep models on tabular cryptocurrency data, and ignoring that would be poor practice.

The reason for including XGBoost when the base paper did not is precisely that the base paper did not. Its comparison set covered Logistic Regression, SVM and Random Forest; gradient boosting, the strongest classical family for this data type, was omitted. Adding it makes our comparison harder to win rather than easier, and a panel respects a team that competes against the strongest available alternative.

Logistic Regression remains in our baseline set because it is the same simple linear comparator the base paper used. Retaining it preserves comparability with the published work

and establishes a performance floor. If a linear model on identical features performs close to our deep model, we are obliged to report that.

If pushed on why deep learning at all: we do not assume it is better, we test it. LSTM, GRU, XGBoost, Random Forest, Logistic Regression and a simple volatility-threshold rule all run on identical features under identical validation. If a simpler model wins, we report that. The base paper found LSTM competitive at an F1 of roughly 0.70; our task is to establish whether that holds on a harder target against a stronger comparison set.

9. Feature Engineering: Base Paper Versus Ours

In the base paper, the authors deliberately avoided conventional price-based variables, arguing that these had already received extensive scrutiny, and instead engineered features from three non-conventional families: blockchain network metrics such as hash rate volatility and large-transaction counts, social sentiment scores, and regulatory event signals. SHAP was used to quantify each family’s contribution, and blockchain-native features were found to supply over a third of the predictive power. SMOTE was applied to address class imbalance.

In BOLT we retain the insight that on-chain features carry genuine predictive signal, and extend the feature space in four directions.

Table 5. Feature families in BOLT and their relationship to the base paper.

Feature family	Examples	Relationship to the base paper
Market / technical	Realised volatility across multiple windows, volume anomaly z-scores, drawdown depth and duration, RSI, momentum	Added back, the base paper excluded these; we test whether they add value alongside on-chain signals
On-chain	Exchange net inflow, whale wallet concentration, active addresses, stablecoin flows	Retained and extended from the base paper
Sentiment	Headline polarity and intensity from finance-domain language models, strictly point-in-time	Retained, with a stricter timestamp guard
Cross-asset contagion	/ Rolling correlation, lead-lag structure, network centrality across assets	New, the base paper models a single asset; this addresses the authors’ own stated future work

One methodological difference deserves explicit statement. We enforce a strict point-in-time constraint, under which a feature value on a given date may only use information publicly available on or before that date. Sentiment leakage, in which text published after a market move is used to “predict” it, is the most common silent failure in this literature. We treat this as a correctness requirement rather than a best-effort target.

On imbalance handling, the base paper used SMOTE and reported a nineteen per cent improvement in recall. We use class weighting and threshold tuning on the precision-recall curve, and will compare against SMOTE as an ablation, because synthetic oversampling of overlapping time-series windows can create near-duplicate examples that leak between folds. This is worth testing rather than assuming.

10. Can We Use an API? If So, Where?

Yes. APIs are used at three distinct points in the pipeline, and it is worth separating them clearly.

The first and principal use is data ingestion: the CoinGecko API for market data, the Binance public API for historical candlesticks, the Etherscan API and Dune Analytics for on-chain flows, CryptoPanic and news RSS feeds for headlines, and the Reddit public API for community text. All provide free tiers sufficient for our data volumes. Collection is scheduled and cached locally, so the frozen dataset does not depend on continued API availability.

The second use is sentiment scoring, through pre-trained finance-domain language models accessed via the Hugging Face library and run locally at inference only. We neither train nor fine-tune a language model; that is outside our scope and unnecessary.

The third use is blockchain interaction, through Web3 RPC endpoints on the Polygon Amoy test network, to submit and read prediction commitments from our smart contract.

It is equally important to state where APIs are not used. No external service is called to produce the prediction itself; the model runs locally on our own data. A system whose output depended on a third-party black-box API could not be independently reproduced, which would defeat the entire purpose of the commitment layer.

11. How Our Model Is Better, and Which Gap It Fills

The gap should be stated as five specific, checkable absences in the base paper rather than as a vague claim of improvement.

Table 6. Gaps in the base paper and how BOLT addresses each.

No.	Gap in the base paper	What BOLT does
G1	Target is a technical chart pattern (pin-bar reversal) rather than a severe market decline	Predicts a drawdown beyond a defined threshold within a forward horizon, an economically meaningful event
G2	Single asset (Bitcoin); no cross-asset structure. This is the authors' own stated future work	Multi-asset, with correlation, lead-lag and contagion features
G3	Comparison set omits gradient boosting and GRU	Benchmarks LSTM, GRU, XGBoost, Random Forest, Logistic Regression and a volatility-threshold rule on identical features
G4	SHAP applied, but explanations never tested for stability across separate episodes	Measures whether the same drivers recur across independent crisis episodes, a reportable finding in either direction
G5	Results rest on a private backtest that no reader can verify	Every prediction is hashed and committed on-chain before the outcome is known, making the track record independently auditable

G5 is our strongest and most original claim. Every predictive paper in this field, the base paper included, asks the reader to trust a backtest that could in principle have been tuned

after the fact, and nothing in the published record distinguishes genuine foresight from hindsight fitting. A cryptographic commitment on a public ledger is the only mechanism that resolves this, and it constitutes a methodological contribution that better models alone cannot supply.

G4 is our strongest research question. Whether the drivers of crash risk remain the same across different crises, or whether every crisis carries its own signature, is genuinely unanswered, and it yields a publishable result in either direction.

12. Answers to Have Memorised

What is your base paper?

Ke et al., 2025, “Early warning of cryptocurrency reversal risks via multi-source data,” Finance Research Letters. An empirical research article in a peer-reviewed Elsevier journal, not a review and not a survey.

What model and what accuracy?

An LSTM, compared against Logistic Regression, SVM and Random Forest. It reports an F1 of 0.703, with rolling-window F1 between 0.64 and 0.72. SHAP analysis showed blockchain features contributing over a third of the predictive power.

Which market?

Hyperliquid, Chainlink, Dai, Uniswap, Aster, Bitcoin, Ethereum, Tether, BNB, and XRP as a single asset, predicting pin-bar reversal events.

How is yours different?

A different target, a severe drawdown rather than a chart pattern. Multi-asset coverage with contagion features. A stronger comparison set including XGBoost and GRU, both of which they omitted. Explanations tested for stability across crises, which nobody has done. And every prediction committed on-chain before the outcome, so our record is auditable rather than merely claimed.

Why is your dataset trustworthy?

Every raw observation comes from a public API or directly from the public blockchain. Nothing is private, nothing is purchased, and nothing is fabricated. The whole dataset regenerates from our scripts, so anyone can rebuild it and verify it.

Why LSTM and XGBoost?

The LSTM reads the multi-day build-up pattern; XGBoost handles the engineered tabular features and pairs well with SHAP. The base paper omitted gradient boosting, and adding it makes our comparison harder to win rather than easier.